LESSON 3.9

Equivalent Ratios

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can find and check equivalent ratios using multiplication and division.

WHY THIS MATTERS Chef Montoya is preparing her famous tomato soup for a school banquet. Her original recipe serves 8 people, but tonight she needs to serve 32 guests. She needs your help to scale the recipe without changing the flavor!

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Equivalent Ratios *Spanish: Razones equivalentes*	Ratios that compare quantities in the same relationship.	2:3 = 4:6 because both parts were multiplied by 2.
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	3 red apples : 5 green apples
Rate *Spanish: Tasa*	A ratio comparing two amounts with different units, like miles per hour.	60 miles per 1 hour
Proportion *Spanish: Proporción*	A math sentence saying two ratios are equal.	2/3 = 4/6
Simplify *Spanish: Simplificar*	To write a ratio with smaller numbers that still compares the same way. 6:8 and 3:4 are the same ratio.	12:18 → divide both by 6 → 2:3
Ratio table *Spanish: Tabla de razones*	A table of equal ratios that shows a pattern.	Cups of juice: 2, 4, 6 | Cups of water: 3, 6, 9

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

Chef Montoya's soup serves 8 people, but she needs to serve 32.

1. I find the scale factor: 32 divided by 8 = 4.
2. The recipe uses 3 cups of tomatoes to 2 cups of broth, a 3:2 ratio.
3. I multiply both by 4: 3 x 4 = 12 cups of tomatoes, and 2 x 4 = 8 cups of broth.
4. So 3:2 and 12:8 are equivalent ratios, and the flavor stays the same.

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A ratio is 2:5. We want to make it bigger using a scale factor of 3.

1. _____
2. _____
3. _____
4. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- I multiplied _____ and _____ by the same number, _____, so the ratio _____ is equivalent to 3:2.

Word bank Equivalent Ratios | Ratio | Rate | Proportion | Simplify | Ratio table | multiply | same number | equivalent | scale | tomatoes | broth

WATCH OUT!

A common mistake

A common mistake in Equivalent Ratios is ADDING the scale factor to one part instead of MULTIPLYING both parts by it. For example, scaling 2:5 by a factor of 4 should give 8:20 (2×4 and 5×4)

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Equivalent Ratios

Level 0 · Most support · every hint printed

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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1 **Circle the letter of the best answer. Show how you know.**

Which ratio is equivalent to 2:5?

(A) 2:10 (B) 3:6

(C) 4:10 (D) 5:2

STEPS TO TRY

1. Equivalent ratios come from multiplying BOTH parts of 2:5 by the same number.
2. For each choice, ask: what did the 2 become, and what did the 5 become? Both must be multiplied by the same amount.
3. Try multiplying both parts of 2:5 by 2 and see what ratio appears. Then look for that pattern among the choices.

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

A recipe uses 4 eggs for every 6 cups of flour. Which ratio is equivalent?

(A) 2:3
(B) 4:3
(C) 6:4
(D) 8:10

STEPS TO TRY

1. You can scale a ratio DOWN too — divide both parts by the same number.
2. Look for a number that divides evenly into both 4 and 6, then divide each part by it.
3. Both 4 and 6 are even, so divide each by 2 and see what simpler ratio you get.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Which ratio is NOT equivalent to 6:9?

(A) 2:3
(B) 3:5
(C) 12:18
(D) 18:27

STEPS TO TRY

1. First simplify 6:9 to its simplest form — that becomes your measuring stick.
2. Simplify each choice too. The odd one out is the choice that does NOT match 6:9's simplest form.
3. Divide both parts of 6:9 by 3 to get its simplest form. Then simplify each answer choice and compare.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

What is the simplest form of the ratio 15:25?

(A) 1:5
(B) 3:5
(C) 5:3
(D) 15:25

STEPS TO TRY

1. Simplest form means both numbers have been divided by their greatest common factor.
2. Find the biggest number that divides evenly into both 15 and 25, then divide each part by it.
3. Both 15 and 25 end in 5, so try dividing each by 5 — and check that nothing bigger divides both.

SHOW YOUR WORK

EXPLAIN YOUR THINKING

How did you find the missing values in Chef Montoya's table? How can you PROVE two of your rows are equivalent ratios?

SENTENCE STARTER I multiplied ___ and ___ by the same number, ___, so the ratio ___ is equivalent to 3:2.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help